SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Questions

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)*

QUESTIONS: 5 **Max Marks: 50** **Duration: 60 Minutes**

Annexure A

Scenario-Based Questions

Code of Conduct:

*I **M. SHAKTHI (192312538)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: M. SHAKTHI

Question 1: Dental X-Ray Enhancement and Cavity Region Extraction

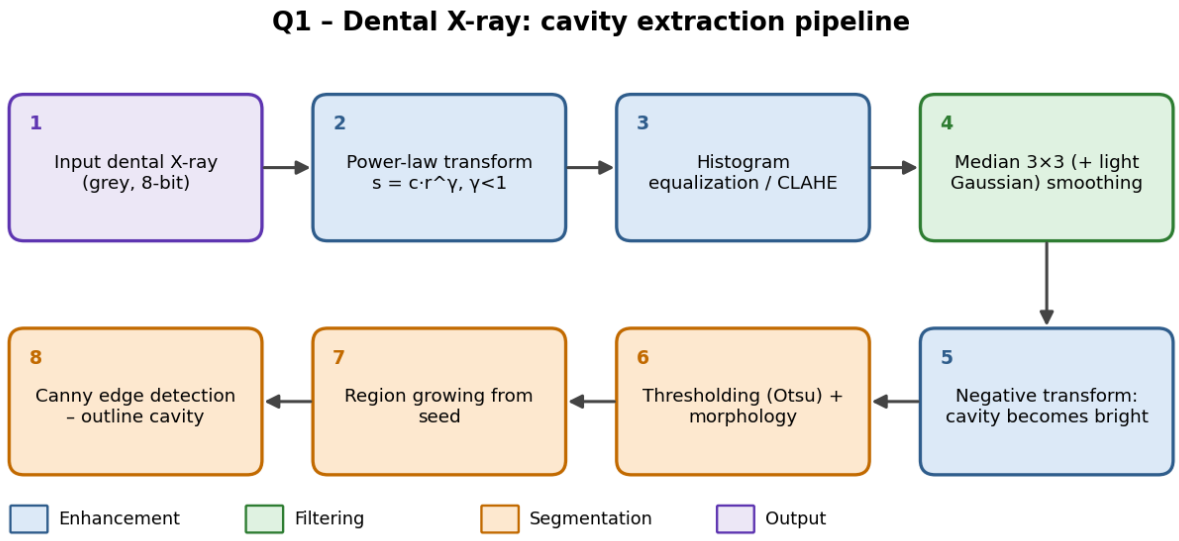
Scenario Understanding

Dental radiographs are dark, low in contrast and corrupted by quantum (Poisson) noise. Early cavities are small radiolucent (darker) patches that blend with enamel and dentine. The goal is to make these patches visible, remove noise without softening their borders, extract the suspected region and outline it.

Key Issues Identified

- **Low brightness and contrast** – most grey levels are crowded in a narrow dark band, so a cavity and its surrounding dentine have almost the same intensity.
- **Quantum noise** – grainy, random variation that can create false small dark spots or hide real ones.
- **Overlapping structures** – enamel, dentine and neighbouring teeth overlap, so a single global rule is risky; the region of interest must be isolated.
- **Boundary sensitivity** – the cavity edge is the clinically important feature, so any smoothing must be edge-preserving.

Processing Pipeline



Step-by-Step Processing and Justification

Step 1 – Grey-level transformation (power-law / log)

Apply $s = c \cdot r^{\gamma}$ with $\gamma \approx 0.4$–0.6 (or $s = c \cdot \log(1+r)$). For $\gamma < 1$ the transform stretches the dark input range into a wider output range, so the hidden detail inside the dark image becomes visible, while bright enamel is compressed so it does not saturate. This is chosen because the whole image is dark; a linear stretch alone would not lift the low-intensity cavity region enough.

Step 2 – Histogram equalization (global or CLAHE)

Compute the cumulative histogram and map $s_k = (L-1) \cdot \Sigma\, p_r(r_j)$ so that grey levels spread over the full range, which raises contrast between the cavity and the surrounding dentine. Because plain equalization also amplifies noise, Contrast-Limited Adaptive HE (clip limit ≈ 2, 8×8 tiles) is preferred when local contrast differs across the tooth.

Step 3 – Spatial smoothing (median 3×3, optional Gaussian σ = 1)

The equalization step has boosted the noise, so it is filtered now. A median filter replaces each pixel by the median of its neighbourhood: it removes isolated noisy pixels but keeps step edges sharp. A plain mean filter would blur the cavity boundary and shrink small lesions, so it is rejected; a light Gaussian is used only to remove the remaining fine grain.

Step 4 – Negative transformation

$s = (L-1) - r$ turns the dark cavity into a bright object on a darker background. This makes the later steps simpler (object = high intensity) and also presents the image in the way many clinicians read lesions. It is a one-to-one point operation, so no information is lost.

Step 5 – Thresholding

Mask the tooth area first so bone and background do not affect the result, then apply Otsu's method, which selects the threshold T that maximises the between-class variance $\sigma_B^2(T)$ of the tooth-region histogram. Pixels above T form the candidate cavity map. Small objects are removed by a morphological opening and by an area filter, and holes are closed, giving a clean binary mask of suspected cavities.

Step 6 – Region growing

A seed is placed inside each candidate (centroid of the component, or selected by the dentist). Neighbouring 8-connected pixels are added while $|I(x,y) - \text{mean(region)}| \leq T_g$ (about 10–15 grey levels). Growth stops when no neighbour satisfies the rule. This refines the rough threshold result into a connected, homogeneous region that follows the true lesion extent and avoids isolated false pixels.

Step 7 – Edge detection (Canny) and outlining

Canny smooths the grown region, finds gradient magnitude and direction with Sobel masks, applies non-maximum suppression to thin the edges and uses hysteresis thresholds to keep strong edges and linked weak edges. The resulting one-pixel-wide contour is overlaid in colour on the original radiograph so the dentist can see the suspected cavity boundary.

Conclusion – Order of Operations

Sequence: enhance (power-law → histogram equalization) → denoise (median) → make the cavity the foreground (negative) → segment (Otsu threshold → region growing) → outline (Canny).
Enhancement is placed before segmentation because thresholding fails on a low-contrast histogram, and filtering is placed after equalization because equalization increases noise. The median filter and Canny edge detector are chosen specifically because they protect the cavity boundary.

Question 2: Night-Time Vehicle Number Plate Region Extraction

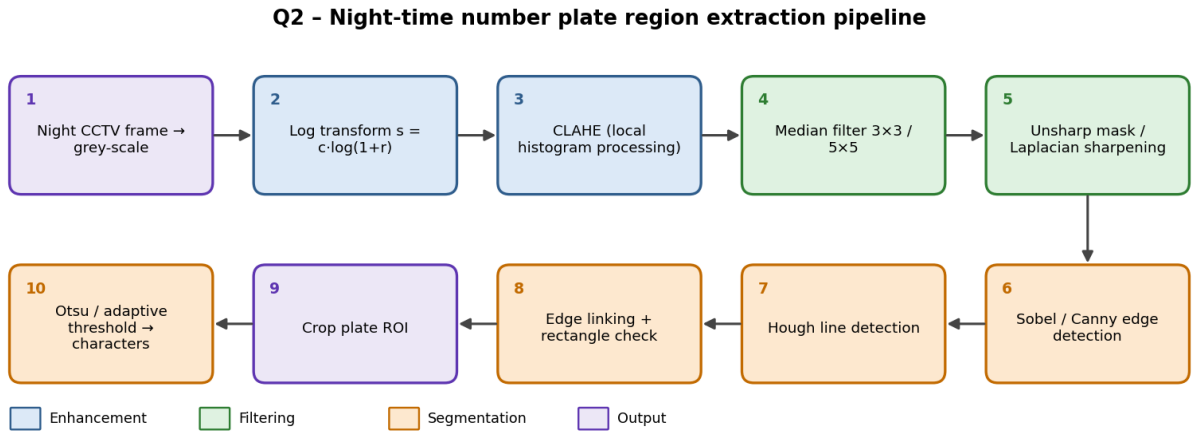
Scenario Understanding

A toll-gate CCTV camera captures vehicles at night. The frames are very dark, headlights create saturated glare, the sensor adds noise, and vehicle motion blurs the characters. The system must locate the rectangular plate and separate its characters for a later recognition stage.

Key Issues Identified

- **Very low brightness** – the plate and body are in the dark end of the histogram.
- **Headlight glare** – a few very bright, saturated regions dominate the histogram and can ruin global enhancement.
- **Sensor noise** – produces false edges that confuse plate detection.
- **Motion blur** – weakens character edges, so sharpening is needed after denoising.
- **Plate geometry** – the plate is a rectangle with a known aspect ratio and a high density of vertical edges (character strokes), which can be used to locate it.

Processing Pipeline



Step-by-Step Processing and Justification

Step 1 – Grey-level transformation (log transform)

Use $s = c \cdot \log(1+r)$, $c = 255 / \log(256)$. The log curve expands dark values strongly and compresses bright values, so it lifts the dark plate while limiting the influence of the glare. Pixels that are almost saturated ($r > 250$) can additionally be clipped or masked so that headlight blobs do not affect the next histogram step.

Step 2 – Histogram processing (CLAHE)

Global histogram equalization is dominated by the large dark background and the glare peaks, so it would over-enhance noise or wash out the plate. CLAHE equalizes each small tile (about 8×8) separately and clips the histogram (clip limit 2–3), which improves local contrast around the plate and prevents the glare area from being over-amplified.

Step 3 – Noise removal (median filter 3×3 or 5×5)

Night sensors produce impulse-type (salt-and-pepper) and speckle noise. The median filter removes it without blurring the edges of the plate border and characters. It is applied before sharpening because sharpening filters amplify noise.

Step 4 – Sharpening (unsharp masking / Laplacian)

Unsharp masking: $g = f + k \cdot (f - f_blur)$, $k \approx 1–2$. Laplacian sharpening: $g = f - \nabla^2 f$ using the 3×3 mask with centre −4 (or −8). Both add the high-frequency (edge) component back to the image, restoring character edges softened by motion blur and by the median filter.

Step 5 – Edge detection (Sobel / Canny)

The gradient magnitude $|\nabla f| = \sqrt{(Gx^2 + Gy^2)}$ finds strong intensity changes. A plate region has many closely spaced vertical edges (character strokes) and long horizontal border edges, so Sobel Gx (vertical edges) followed by horizontal dilation highlights the plate zone while the plain car body gives sparse edges.

Step 6 – Line detection (Hough transform)

Each edge pixel votes in the (ρ, θ) space for lines $\rho = x \cdot \cos\theta + y \cdot \sin\theta$. Peaks near $\theta \approx 0°$ and 90° give the top/bottom and left/right borders of the plate. This is robust to partial breaks in the edges and to small noise, which simple scanning would not tolerate.

Step 7 – Edge linking

Gaps in the plate border caused by glare or noise are bridged by local linking: two edge pixels are joined if their gradient magnitudes and directions are similar ($|\nabla f(x,y) - \nabla f(x_0,y_0)| \leq E$ and $|\alpha(x,y) - \alpha_0| \leq A$), supported by morphological closing. The linked lines are intersected to get four corners and checked against plate rules (aspect ratio about 2:1 to 5:1, minimum area, expected position).

Step 8 – Plate cropping and thresholding

The rectangular region is cropped. Inside it, Otsu's threshold (or an adaptive/Sauvola threshold if illumination is uneven) separates characters from the plate background; the polarity is inverted if characters are lighter than the plate. Connected-component analysis keeps blobs with character-like height and width, giving a clean binary character image for the next recognition stage.

Conclusion – Order of Operations

Sequence: brighten (log) → control glare/local contrast (CLAHE) → denoise (median) → restore edges (unsharp/Laplacian) → locate (edges → Hough lines → edge linking) → binarize (Otsu/adaptive). Each stage prepares the next: enhancement makes edges detectable, denoising prevents false edges, sharpening strengthens true edges, and line detection with linking uses the plate's rectangular geometry to isolate it from the vehicle body.

Question 3: Textile Fabric Defect Detection Using Frequency-Domain Processing

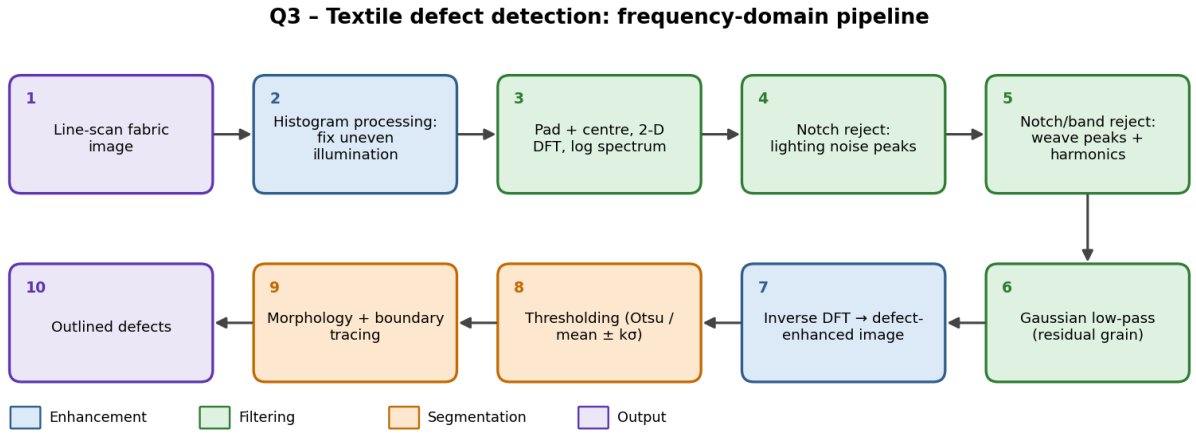
Scenario Understanding

A line-scan camera images woven fabric. A defect-free weave is highly periodic; holes, broken threads, oil stains and knots break that periodicity. The images also contain periodic noise from machine lighting and uneven illumination across the roll width.

Key Issues Identified

- **Regular weave texture** – appears as sharp, regularly spaced peaks (with harmonics) in the Fourier spectrum.
- **Periodic lighting noise** – also appears as isolated symmetric spikes in the spectrum.
- **Uneven illumination** – a very slow brightness change across the width; it is mainly a low-frequency/histogram problem and must be corrected before thresholding.
- **Defects** – local, non-periodic; their energy is spread over many frequencies, so they survive when the periodic peaks are removed.

Processing Pipeline



Step-by-Step Processing and Justification

Step 1 – Histogram processing for uneven illumination

The image is divided into column blocks (or processed with CLAHE) and each block's histogram is equalized or matched to a reference histogram taken from a defect-free, well-lit strip. This brings the average brightness and contrast of the left, centre and right part of the roll to the same level. It is done first so that the illumination gradient does not leak into the low-frequency part of the spectrum or into the final threshold.

Step 2 – Forward 2-D DFT

The image is zero-padded to avoid wrap-around error and multiplied by $(-1)^{(x+y)}$ so that the origin of the spectrum is at the centre. $F(u,v) = \Sigma\Sigma\, f(x,y)\cdot e^{(-j2\pi(ux/M + vy/N))}$ is computed with the FFT. The log-magnitude spectrum, $\log(1+|F|)$, is displayed to locate the bright peaks that belong to the weave and to the lighting noise.

Step 3 – Notch reject filter for lighting noise

A Butterworth notch-reject filter $H(u,v) = 1 / (1 + [D_0^2 / (D_1 \cdot D_2)]^n)$ is placed at each symmetric peak pair $(\pm u_0, \pm v_0)$, where D_1 and D_2 are the distances to the two peaks of a pair. It removes only the

narrow frequency bands occupied by the machine-lighting stripes and leaves everything else untouched. Butterworth (order 2) is preferred to an ideal notch because the ideal filter causes ringing.

Step 4 – Notch / band-reject filter for the weave texture

The same type of notch filters (or a band-reject filter if the peaks lie on a ring) are applied at the fundamental weave frequencies and their harmonics. After this the regular pattern is removed from the image, but the defects, which are not periodic, remain. This is the key step that makes the defect "stand out".

Step 5 – Gaussian low-pass filtering (optional)

A Gaussian low-pass filter $H = e^{(-D^2/2D_0^2)}$ removes any remaining fine grain or high-frequency noise without ringing. If the edges of thin defects such as broken threads need to be emphasised, a mild high-frequency-emphasis (high-pass) filter can be used instead, because it keeps the sharp boundaries of defects.

Step 6 – Inverse DFT

$g(x,y) = Re\{ IDFT[H(u,v)\cdot F(u,v)] \}\cdot(-1)^{(x+y)}$, followed by cropping the padded area. The result is an image in which the background is almost flat and uniform and only the defects have intensity deviation.

Step 7 – Thresholding

Because the background is now flat, a simple global threshold works: Otsu's method, or $T = \mu \pm k\cdot\sigma$ ($k \approx 3$) on the residual image, so that bright and dark deviations are both captured. Morphological opening and closing remove tiny false responses and fill small gaps inside a defect.

Step 8 – Boundary detection and defect description

Boundary extraction $\beta(A) = A - (A \ominus B)$ or contour tracing (Moore boundary following) / Canny gives the outline of every defective region. Area, elongation and mean intensity of each region help to classify it: a hole is dark and compact, a broken thread is thin and elongated, an oil stain is large and irregular with soft contrast, a knot is a small bright blob.

Conclusion – Order of Operations

Sequence: correct illumination (histogram processing) → DFT → notch/band-reject removal of lighting noise and weave → inverse DFT → threshold → outline defects. Frequency-domain processing is preferred because the interfering signals (weave and lighting noise) are periodic, so each is concentrated in a few narrow spikes in the spectrum that can be deleted precisely, leaving the non-periodic defects intact. In the spatial domain, removing a repeating texture needs large averaging or subtraction masks that blur or erase small defects and cannot separate them from the pattern; in addition, FFT-based filtering ($O(MN \log MN)$) is far faster than large-kernel spatial convolution on long line-scan images and works on the whole width at once.

Question 4: Blood Cell Segmentation in Microscopic Images

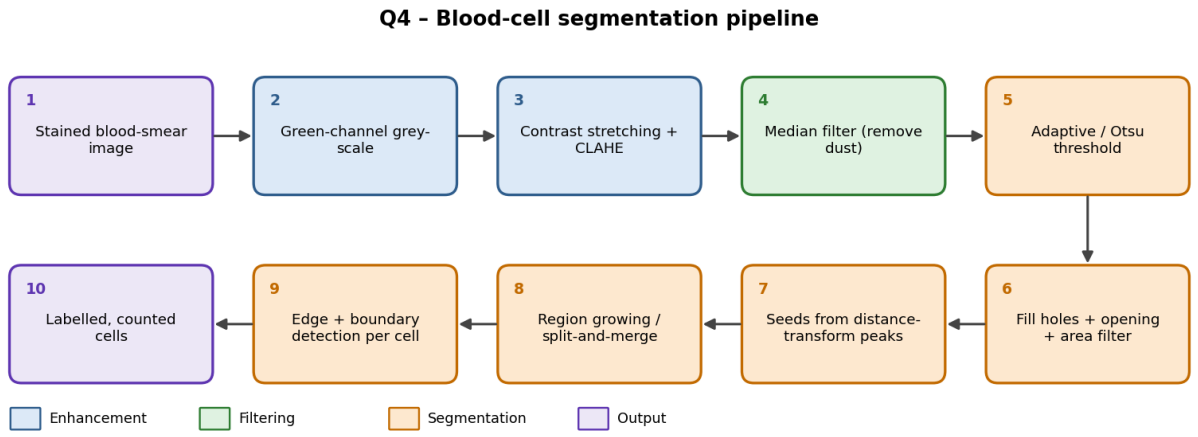
Scenario Understanding

Stained blood-smear images are used to count red (RBC) and white (WBC) cells. The images show uneven staining, weak contrast between cells and background, touching or overlapping cells and small dust particles. Each cell must be separated and outlined so that it can be counted correctly.

Key Issues Identified

- **Uneven staining and poor contrast** – a single global threshold would merge or miss cells.
- **Dust and small particles** – produce false small objects that would be counted as cells.
- **Touching / overlapping cells** – thresholding makes them one blob, which gives a wrong count.
- **Boundary accuracy** – cell size and shape are diagnostically important, so edges must be preserved.

Processing Pipeline



Step-by-Step Processing and Justification

Step 1 – Grey-scale conversion

The colour smear image is converted to grey-scale using the green channel (or saturation), because stained cells absorb green light strongly and therefore show the highest cell-to-background contrast in this channel.

Step 2 – Contrast enhancement and histogram processing

Contrast stretching $s = (r - r_min)\cdot(L-1)/(r_max - r_min)$ with percentile limits (1 %–99 %) uses the full grey range. This is followed by histogram equalization/CLAHE, which equalizes tiles separately and so reduces the effect of uneven staining across the slide. The histogram becomes more bimodal (background and cells), which helps the thresholding step.

Step 3 – Spatial filtering (median filter 3×3 / 5×5)

Dust particles are small, isolated, impulse-like specks. The median filter removes them completely while keeping the cell boundaries sharp. A mean or Gaussian filter would only smear dust into grey blobs and soften the cell edges, so it is not used as the main filter.

Step 4 – Global or adaptive thresholding

Otsu's global threshold is used if the background is now uniform. Because staining is uneven, an adaptive (local) threshold $T(x,y)$ = local mean (or Gaussian-weighted mean) − C over a window (about

31×31) is preferred, as it adapts to the local background. The binary image is then cleaned: holes (the pale centre of an RBC) are filled, a morphological opening removes remaining specks, and objects smaller than a minimum area are discarded.

Step 5 – Region-based segmentation of touching cells

Thresholding gives touching cells as one connected blob. To split them, the distance transform of the binary image is computed; its local maxima give one seed per cell centre. Region growing starts from each seed and adds neighbouring pixels satisfying a similarity rule until fronts from different seeds meet, so each cell gets its own label. Alternatively, region splitting and merging can be used: the image is recursively split into quadrants until each satisfies the predicate P (e.g. $\sigma < T$), then adjacent regions whose union still satisfies P are merged.

Step 6 – Edge detection and boundary detection

A Canny/Sobel edge detector applied to each labelled region, or boundary extraction $\beta(A) = A - (A \ominus B)$, gives a one-pixel contour per cell. Contours are drawn over the original image and connected-component labelling gives the count. Size and intensity features (large cell with dark nucleus = WBC, smaller pale-centre cell = RBC) are used to separate cell types.

Conclusion – Order of Operations

Sequence: enhance contrast (green channel → stretching → CLAHE) → remove dust (median) → separate cells from background (adaptive/Otsu threshold + morphology) → split touching cells (seeded region growing or split-and-merge) → outline (edge/boundary detection) → count. The thresholding step supplies the cell mask, but only the region-based step can give an accurate count because it converts merged blobs into individual cell regions, and edge detection finally supplies the accurate cell outlines for size and shape measurement.

Question 5: Text Extraction from Degraded Historical Documents

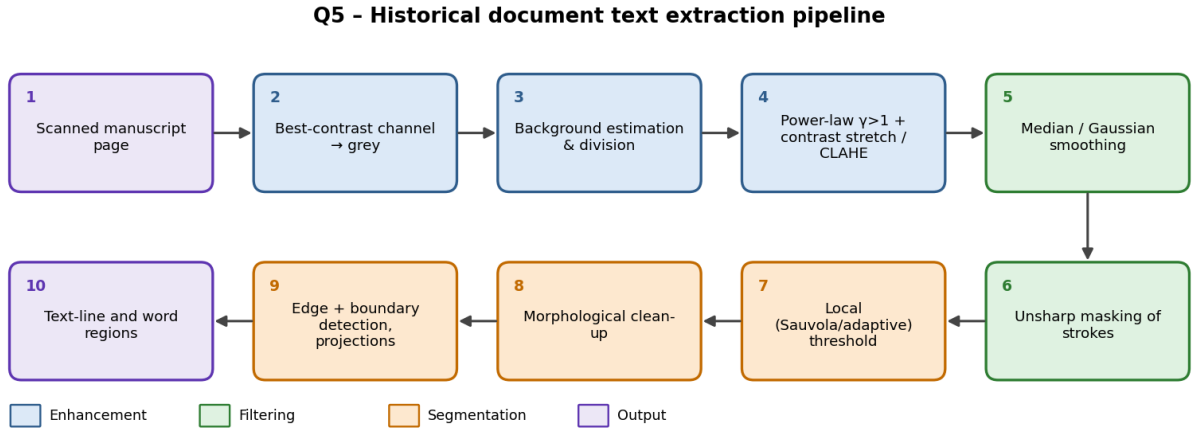
Scenario Understanding

A state archive is digitising old handwritten manuscripts. The scans have yellowed paper, ink bleed-through from the reverse side, stains, faded strokes and uneven lighting. Clean text regions must be extracted before handwriting recognition.

Key Issues Identified

- **Faded strokes** – ink grey level is close to the paper level, so contrast must be raised.
- **Bleed-through and stains** – intermediate grey regions that can be wrongly treated as text.
- **Uneven lighting and yellowed paper** – background brightness varies across the page, so one global threshold cannot work.
- **Handwriting** – strokes are thin and irregular; smoothing must not erase them.

Processing Pipeline



Step-by-Step Processing and Justification

Step 1 – Grey-scale conversion and background correction

The colour page is converted using the channel with best ink-to-paper contrast (typically the red or green channel, because yellowed paper absorbs blue). The slowly varying background is estimated with a large median or morphological closing (kernel ≈ 51×51) and the page is divided by it: g = f / background. This flattens uneven lighting and removes large soft stains before enhancement.

Step 2 – Grey-level transformation and histogram processing

A power-law transform s = c·r^γ with γ ≈ 1.5–2.5 darkens mid-tone (faded) strokes while the near-white background stays bright, which increases the faded-ink/background gap. Contrast stretching and CLAHE (clip ≈ 2) then use the full grey range locally. The faint reverse-side bleed-through stays at an intermediate level, which makes it separable in the thresholding step.

Step 3 – Smoothing

A median filter (3×3) removes speckle and small stains; a Gaussian filter (σ ≈ 1) can smooth paper grain. Both are mild in size so that thin pen strokes are not erased. A bilateral filter may be used when edge preservation is critical.

Step 4 – Sharpening (unsharp masking)

$g = f + k \cdot (f - f_{blur})$, $k \approx 1.5$, re-strengthens the pen strokes weakened by smoothing and by fading, so the stroke edges become more distinct from the background. It is applied after smoothing so that noise is not amplified.

Step 5 – Global (Otsu) vs local (adaptive) thresholding

Otsu (global): one threshold for the whole page from the histogram; fast and good for clean, uniformly lit scans with a bimodal histogram. In this scenario the histogram is not bimodal (stains, bleed-through, shadows), so dark stains become false text and faded strokes in bright areas are lost. **Local/adaptive (Sauvola or Gaussian adaptive):** $T(x,y) = m(x,y) \cdot [1 + k(s(x,y)/R - 1)]$ computed in a window of 25–51 px, with local mean m and standard deviation s. It follows the changing background and local contrast, so it keeps faded strokes and suppresses stains and bleed-through. **Decision:** local (Sauvola/adaptive) thresholding is the more suitable method and is applied.

Step 6 – Morphological clean-up

Connected components smaller than a minimum area (specks) are removed and a small closing joins strokes broken by fading. This gives a clean binary text image.

Step 7 – Edge detection and boundary detection for text lines and words

Canny/Sobel edges and contour tracing locate stroke boundaries. For text lines, the binary image is dilated with a wide horizontal element (about 1×40 px) so that letters merge into line blobs; contour tracing / the horizontal projection profile (row sums, valleys = line gaps) gives each line's bounding box. Inside a line, a smaller horizontal dilation or the vertical projection profile (gaps larger than the typical letter gap) separates word regions. These boxes are passed to the handwriting recogniser.

Conclusion – Order of Operations

Sequence: correct the background → raise stroke contrast ($\gamma > 1$, CLAHE) → denoise and de-stain (median/Gaussian) → sharpen strokes → local threshold (Sauvola/adaptive, chosen over Otsu) → clean up → find lines and words by boundary detection. Each decision follows the degradation: background division for uneven light, gamma and CLAHE for faded ink, mild smoothing for stains, local thresholding for varying background, and horizontal dilation/projection for line and word grouping.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand